

Monte Carlo dosimetry of organ doses from a sweeping-beam total body irradiation technique: feasibility and first results

Levi Burns^{1,2}, Tony Teke^{3,4}, I. Antoniu Popescu^{1,2}, Cheryl Duzenli^{1,2}

¹ Department of Physics & Astronomy, University of British Columbia, Vancouver BC, Canada

² Medical Physics, BC Cancer (Vancouver Centre), Vancouver BC, Canada

³ Department of Computer Science, Mathematics, Physics & Statistics, University of British Columbia (Okanagan), Kelowna BC, Canada

⁴ Medical Physics, BC Cancer (Southern Interior), Kelowna BC, Canada
levi.burns@bccancer.bc.ca

Abstract. Total body irradiation (TBI) is a radiation treatment often purposed to suppress the immune system prior to a bone marrow transplant. Several toxicities can arise in TBI, and high-quality dose volume data for organs at risk are required if one is considering any change from a well-established technique. We present a novel Monte Carlo (MC) dosimetry technique to acquire this data based on our current TBI technique that accounts for a sweeping-beam Cobalt-60 delivery, a stationary flattening filter, patient-specific lung compensators, and two patient treatment positions. For each patient, a virtual MC phantom is created including the planning CT image in each treatment position (supine and prone). The results from dose simulations on each phantom were summed together geometrically with a deformable registration tool. Dose volume statistics for lungs, liver, thyroid, and kidneys are obtained. The preliminary results of a retrospective study using this technique on patients who have received TBI at our clinic indicate that, for a total body prescription dose of 12 Gy $\pm$ 10%, the mean body dose ranged from 11.19 to 12.15 Gy with smaller patients receiving lower mean body doses than larger patients. The mean dose delivered to the thyroid was the highest of the contoured organs receiving up to 12.84 Gy, and the lung doses were the most heterogeneous, with standard deviations up to 0.73 Gy in individual patients. This high-quality dose data shows promise for use in both routine quality assurance of our current technique, and to provide baseline data for development of a new technique. The technique could also be adapted to TBI techniques at other clinics that include compensators, flattening filters, moving beams, and/or multiple treatment positions.

Keywords: total body irradiation, Monte Carlo, dosimetry, Cobalt, sweeping beam, organ dose

1 Introduction

Total body irradiation (TBI) entails the delivery of a uniform radiation dose to the entire patient. Its use is often part of a conditioning regimen to prevent the onset of graft-versus-host disease (GvHD) in patients who will receive a bone marrow transplant (BMT) [1, 2]. A recent survey across Canadian cancer centres estimates that TBI is received by roughly 400 people nation-wide per year [3]. This study, in conjunction with an international survey of a similar nature [4], illustrates a large amount of diversity in how TBI is delivered, with approaches including translating treatment couches, parallel-opposed pair treatments, and volumetric modulated arc therapy (VMAT). Several toxicities are reported from TBI, the most important of which is interstitial pneumonitis which can be fatal. It is important to acknowledge the difficulty in discerning whether toxicities following TBI are due to the radiation treatment or other aspects of the patient's care, such as concurrent chemotherapy or the BMT itself [5], although precautions may still be taken during TBI to reduce the incidence of these effects.

Our clinic currently uses a sweeping-beam Cobalt-60 technique to deliver TBI. While our treatment is effective, we are considering an upgrade to a linac-based method that will improve the patient experience and open the door to possible customization of organ doses through intensity modulation. To ensure continuity of clinical care, it is desirable to establish dosimetric equivalence of any new technique with the current technique, prior to implementation. We thus require accurate, high-quality organ dose data from the current treatment which has, until recently, been unavailable for individual patients. The gold standard for obtaining this data would be Monte Carlo (MC) dosimetry. Some efforts at implementing MC procedures into the TBI treatment planning pipeline have been described in the literature, primarily to evaluate the homogeneity of the dose distribution and the effectiveness of flattening filters, but organ dose data from TBI has yet to be reported [6-9].

In this paper, we present a technique for TBI dosimetry using Monte Carlo simulations based on our sweeping-beam technique, including a deformable registration to add together doses from two different treatment positions. We demonstrate proof of principle with initial findings for a small retrospective series of patients who have received TBI at our centre.

2 Methods

2.1 Description of TBI technique

Patients in our clinic receive TBI from a repurposed Cobalt-60 unit (Theratron 780C, Atomic Energy of Canada Ltd., Ottawa, Canada). An early version of our technique is described by Sherali and El-Khatib [10]. A schematic of the treatment is given in Figure 1. The patient is positioned at an extended SSD, typically around 160 cm, depending on the size of the patient. The field (70x70 cm² at 160 cm from the source) covers a section of the patient at a given time, and the gantry repeatedly sweeps out a 90-degree angle during treatment to irradiate the entire body. Each fraction is delivered in two parts, with the patient lying in supine and prone positions. Before each treatment, CT image volumes are collected from the patient in both positions, from the level of the

pelvis to the level of the chin. These are used to design patient-specific lead lung compensators for each treatment position for each patient to bring the lung dose down to prescription level. The positions of the lung compensators are verified by a physicist at the first treatment with a digital imaging panel placed underneath the treatment bed. A stationary polystyrene flattening filter is centered beneath the Cobalt source to ensure a uniform dose is delivered to the whole patient.

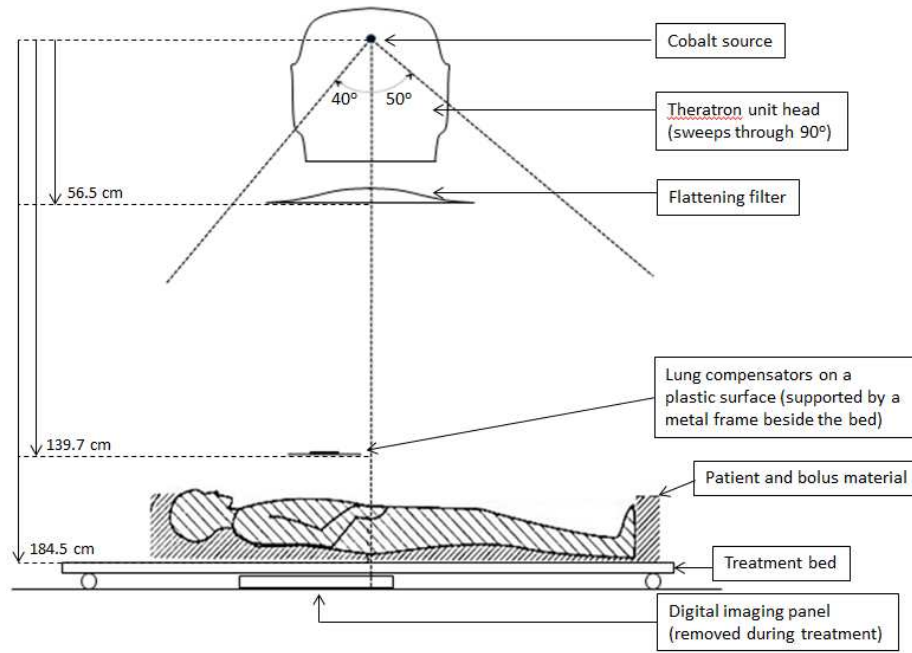


Fig. 1. Schematic of our sweeping-beam total body irradiation technique.

2.2 Monte Carlo phantom production

To run a Monte Carlo simulation, a phantom corresponding to the modelled system must first be created in the simulation software. We have designed a method to produce these phantoms using Python, Matlab, and C++ codes. The phantoms are designed for use with the EGSnrc Monte Carlo simulation software [11, 12]. A complete phantom for a patient is shown in Fig. 2.



Fig. 2. Axial (left) and sagittal (right) slices of a Monte Carlo phantom for a TBI patient in the supine position. The three structures above the patient are the flattening filter, the lung compensators (white), and the plastic tray on which the lung compensators rest. The visualization software used is unable to display variable resolutions, although this depiction remains useful to conceptualize the geometry of the MC phantoms used. In the simulations, these structures are positioned at appropriate distances as indicated in Fig. 1.

The patient CT image is converted to a MC phantom using the software CTcreate [13] in the EGSnrc platform. The lung blocks for the patient treatment are retrieved and converted to a phantom with an in-house Matlab script. A phantom for the polystyrene flattening filter was created by taking a CT scan of it and using CTcreate. The phantoms of the lung blocks, flattening filter, and the plastic sheet on which the lung blocks rest are appended as new slices above the original patient phantom at a height and position corresponding to the treatment specifications. Variable resolutions are used in the beam axis direction to model the large air gaps between phantom components as single air voxels. This reduces the computational time that would be required to simulate histories through large areas of air voxels. The resolution in the other directions is $0.25 \times 0.25 \text{ cm}^2$ for all phantom components.

2.3 Patient selection and ethics approval

Patient cases selected for this retrospective study had been prescribed a TBI treatment of 1200 cGy in 6 fractions since August 2015. Cases were excluded from the study if the CT images revealed resected organs. For this initial study, both male and female patients with a variety of body sizes were included, including pediatric patients. The study has been approved by our institutional Research Ethics Board.

2.4 Monte Carlo simulations and deformable registration

A model of the Cobalt source and housing was produced by Mora et al [14] and is used in our simulations. Simulations of this source while the beam is sweeping are made possible by Source 21 of DOSXYZnrc [15]. Three billion particle histories are simulated for each treatment position with maximum dose uncertainties below 2% in the body and below 3% in the lungs. The simulations are run in parallel in up to 256 jobs simultaneously (8 GB RAM, 2.00 GHz processors) and take between 60-90 minutes to complete depending on the size of the phantom.

After the simulations are performed, the flattening filter, lung blocks, and plastic compensator tray are removed from the phantom to restore a uniform resolution along the beam direction for analysis. Dose distributions are de-noised with a 3D Savitzky-Golay filter [16].

The doses given by an EGSnrc MC simulation are given in units of Gy per particle incident on a linear accelerator head target [17], which is not the circumstance for our purposes. To scale the relative doses from the simulation to real doses, a calibration measurement was undertaken by irradiating a solid water phantom at TG-51 conditions [18], with an ion chamber dosimeter at the standard reference point. An identical geometry was modelled as an EGSnrc phantom and simulated with two billion particle histories. The computed dose and the measured dose at the calibration point from this experiment were used to scale patient doses to Gy. Our sweeping Monte Carlo source has been validated in water tank measurements in previous work [19] to be described in an upcoming report.

To add the doses from each treatment position in an appropriate manner, the simulated dose files from supine and prone treatments are transferred to MIM Maestro™ v6.6.8 (MIM Software Inc., Cleveland, OH) where a deformable registration tool is available. The deformable registration is intensity-based and fully automatic.

2.5 Analysis

Organ dose data is analyzed in MIM Maestro™ following the deformable registration. The contoured organs are the kidneys, lungs, liver, and thyroid. Additionally, the whole body dose was considered for all patients. In the body contour, because the clinical protocol for CT simulations only covers part of the patient, three axial slices were removed from each of the superior and inferior edges of the body contour, as these doses are missing a scatter dose contribution from the rest of the body not included in the CT image volume. For a similar reason, the arms are removed manually from each body contour because the simulations do not incorporate bolus material that is used in patient positioning, and thus the arms are missing a scatter dose contribution in the simulations in comparison to the actual treatment. Moreover, the different positions of the arms between the supine and prone positions renders this a challenging area for deformable registration of dose distributions.

3 Results

The TBI treatments of 11 patients were simulated and the dose results for organs and the whole body were collected. The supine and prone dose distributions for a patient are shown in Fig. 3 along with the final distribution after deformable registration.

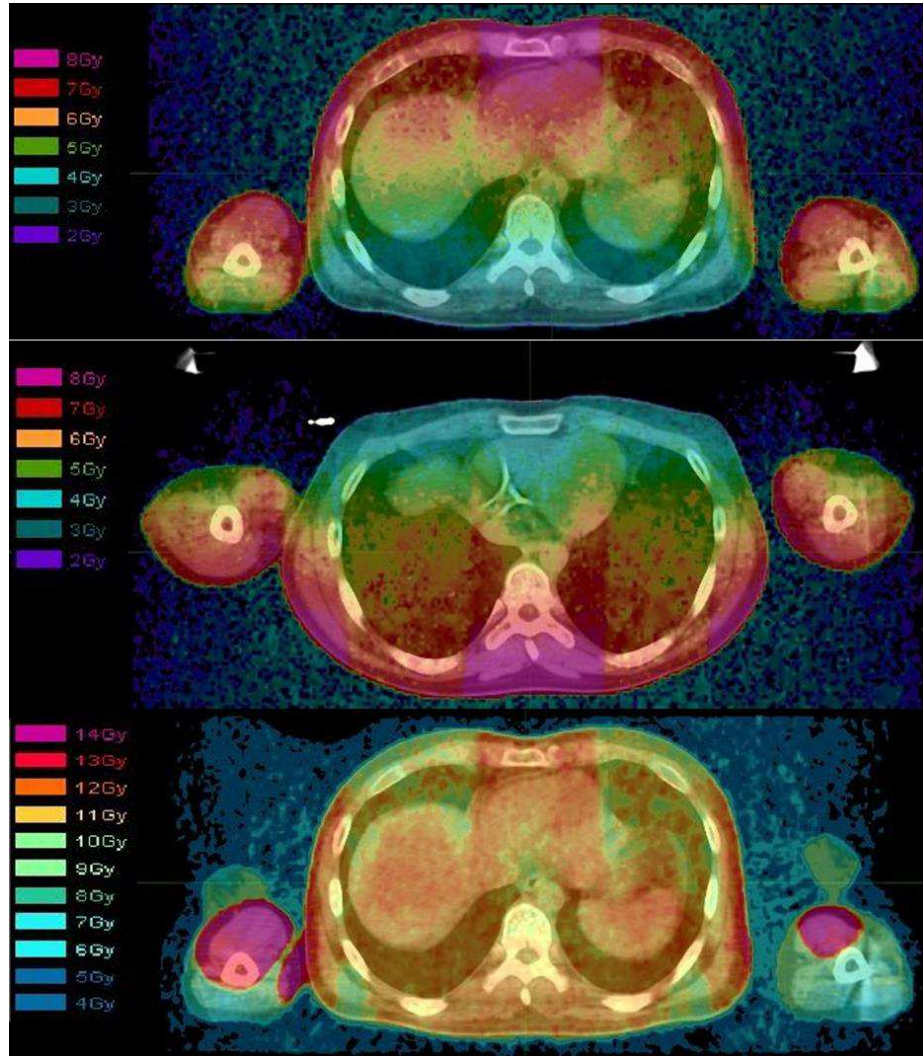


Fig. 3. The simulated dose distribution is shown for the supine (top) and prone (middle) simulations separately, and the final distribution after deformation is projected onto the supine image (bottom). The prone image (middle) is displayed upside-down for comparison purposes with the supine images.

In Fig. 4, the mean body dose is plotted against the size of the patient, as measured by the patient mid-separation depth at the level of the umbilicus. All mean body doses fall within the 12 Gy $\pm$ 10% prescription (range: 11.19-12.15 Gy), but within this constraint, the data suggest that smaller patients receive lower mean body doses.

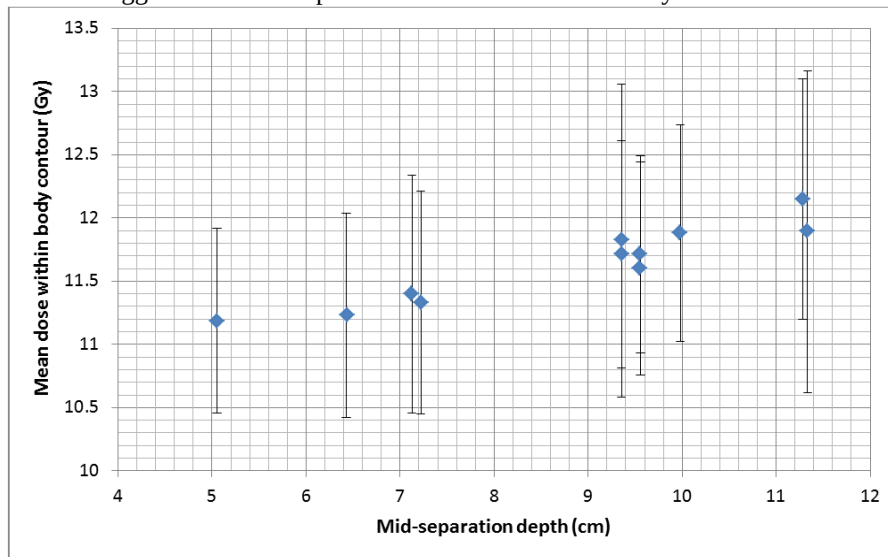


Fig. 4. Mean body dose plotted against patient mid-separation depth at the level of the umbilicus. The error bars represent the standard deviation of dose to voxels within the body contour.

Preliminary organ dose statistics are shown in Table 1, representing the results across the 11 simulated treatments. In addition to this data across the patient population, individual organ doses were examined on a case-by-case basis. For each patient, the doses to the thyroid and kidneys were consistently above the mean body dose, while the doses to the liver and lungs were consistently below, and the lung doses appear to be the most heterogeneous with standard deviations up to 0.73 Gy in one case. A further discussion of these results is being prepared in an upcoming publication.

Table 1. A summary of organ dose statistics for 11 TBI patients. Each mean dose is the average across the patients. The maximum and minimum doses across the patients are also given.

Organ	Mean dose (Gy)	Minimum dose (Gy)	Maximum dose (Gy)
Lung (left)	11.32	10.96	11.89
Lung (right)	11.25	10.73	11.75
Kidney (left)	11.86	11.22	12.52
Kidney (right)	11.86	11.15	12.34
Liver	11.37	10.98	11.73
Thyroid	12.07	11.10	12.84
Body contour	11.63	11.19	12.15

4 Discussion

First and foremost, we have demonstrated the feasibility of employing Monte Carlo dosimetry for a sweeping-beam Cobalt-60 TBI technique incorporating lung blocks, a flattening filter, a moving beam, and two treatment positions. This data will be invaluable for setting clinical dose constraints in the design of a new TBI technique. The Monte Carlo technique has potential to be implemented as a routine quality assurance tool. Further studies are underway to optimize and streamline simulations, and to simulate a diverse patient population and obtain further information about organ doses.

In this study, three billion histories per phantom simulated were used. This number was chosen based on the authors' previous experience with MC simulations. Experiments are currently underway to determine if simulations with a lower number of particle histories per phantom could be used to obtain results of similar quality, in the interest of reducing simulation times.

The technique to produce these phantoms currently requires about 75 minutes of manual work per patient by an expert user. For clinical implementation, this would be an unacceptable amount of time for quality assurance of more common treatments, but it is not unfathomable given that the average number of TBI cases in our clinic is less than one patient per week. There is potential to further streamline the phantom production workflow and this is a current area of focus.

Validating the algorithm for a prone-to-supine deformable registration is a challenging task and is beyond the scope of this work. However, the RegReveal tool in MIM MaestroTM allows for qualitative inspection of local performance of the registration algorithm, and the CT images contain a large amount of bony anatomy to guide the registration, including the entire spine. It should also be noted that the quality of the registration can be improved further by the use of "locks" by adept users of MIM Maestro which allow for iterative deformations of challenging areas while preserving existing deformations in successful areas. Because the placement of locks would vary by user, we have not made any attempt to implement them at this stage in the interest of the reproducibility of our results.

Finally, it can be noted that our technique in producing MC phantoms, where material slices such as plastic filters and lead compensators are appended slice-by-slice at appropriate heights above a patient CT scan, could be applied to TBI techniques at other clinics, or to other treatments where physical objects lie in the path of the beam, with minor adjustments.

5 Conclusions

We have created a method to obtain Monte Carlo organ dose information of our sweeping-beam TBI technique, and it could feasibly be implemented as a routine quality assurance tool for our treatments in the future. Preliminary retrospective results indicate that patient doses are consistent with their prescriptions with some patterns of variation within the clinically acceptable dose limits for each organ.

Acknowledgements

This work was supported in part by an Affiliated Fellowship from the UBC Faculty of Graduate and Postdoctoral Studies.

Conflict of Interest

The authors declare that they have no conflict of interest.

References

1. Hill-Kayser CE, Plastaras JP, Tochner Z, Glatstein E: TBI during BM and SCT: review of the past, discussion of the present and consideration of future directions. *Bone Marrow Transplantation* 46(4), 475-484 (2011).
2. Wills C, Cherian S, Yousef J, Wang K, Mackley HB: Total body irradiation: a practical review. *Appl Rad Oncol* 5(2), 11-17 (2016).
3. Studinski RCN, Fraser DJ, Samant RS, MacPherson MS: current practice in total-body irradiation: results of a Canada-wide survey. *Current Oncology* 24(3) (2017).
4. Geibel S, Miszczyk L, Slosarek K, Moukhtari L, Ciceri F, Esteve J, et al: Extreme heterogeneity of myeloablative total body irradiation techniques in clinical practice. *Cancer* 120(17), 2760-2765 (2014).
5. Kelsey CR, Horwitz ME, Chino JP, Craciunescu O, Steffey B, Folz RJ, et al: Severe pulmonary toxicity after myeloablative conditioning using total body irradiation: an assessment of risk factors. *Int J Radiat Oncol Biol Phys* 81(3), 812-828 (2011).
6. Charakova R, Muntzing K, Krantz M, Hedin E, Hertzman S: Monte Carlo optimization of total body irradiation in a phantom and patient geometry. *Phys Med Biol* 58(8), 2461-2469 (2013).
7. Charakova R, Krantz M: A Monte Carlo evaluation of beam characteristics for total body irradiation at extended treatment distances. *J Appl Clin Med Phys* 15(3), 182-189 (2014).
8. Serban M, Seuntjens J, Roussin E, Alexander A, Tremblay JR, Wierzbicki W: Patient-specific compensation for Co-60 TBI treatments based on Monte Carlo design: A feasibility study. *Phys Med* 32(1), 67-75 (2016).
9. Liu X, Lack D, Rakowski JT, Knill C, Snyder M: Fast Monte Carlo simulation for total body irradiation using a 60Co teletherapy unit. *J Appl Clin Med Phys* 14(3), 133-149 (2013).
10. Sherali H, El-Khatib E: Total body irradiation with a sweeping 60-Cobalt beam. *Int J Radiat Oncol Biol Phys* 33(2), 493-497 (1995).
11. Rogers DWO, Faddegon BA, Ding GX, Ma CM, We J, Mackie TR: BEAM: A Monte Carlo code to simulate radiotherapy treatment units. *Med Phys* 22(5), 503-524 (1995).
12. Kawrakow I: Accurate condensed history Monte Carlo simulation of electron transport. I. EGSnrc, the new EGS4 version. *Med Phys* 27(3), 485-498 (2000).
13. Walters B, Kawrakow I, Rogers DWO: DOSXYZnrc users manual Report PIRS-794. National Research Council of Canada (2017).
14. Mora GM, Maio A, Rogers DWO: Monte Carlo simulation of a typical 60Co therapy source. *Med Phys* 26(11), 2494-2502 (1999).
15. Lobo J, Popescu IA: Two new DOSXYZnrc sources for 4D Monte Carlo simulations of continuously variable beam configurations, with applications to RapidArc, VMAT, TomoTherapy and CyberKnife. *Phys Med Biol* 55(16), 4431-4443 (2010).
16. Kawrakow I: On the de-noising of Monte Carlo calculated dose distributions. *Phys Med Biol* 47(17), 3087-3103 (2002).

17. Popescu IA, Shaw CP, Zavgorodni SF, Beckham WA. Absolute dose calculations for Monte Carlo simulations of radiotherapy beams. *Phys Med Biol* 50(14), 3375-3392 (2005).
18. Almond PR, Biggs PJ, Coursey BM, Hanson WF, Huq MS, Nath R, Rogers DWO: AAPM's TG-51 protocol for clinical reference dosimetry of high-energy photon and electron beams. *Med Phys* 26(9), 1847-1870 (1999).
19. Teke T: Monte Carlo techniques for patient specific verification of complex radiation therapy treatments including TBI, VMAT and SBRT lung. Ph.D thesis (not published), University of British Columbia (2012).